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Language

English ▼

Network Monitoring

In this section, you will utilize the three VPC's with Internet Gateways, Transit Gateway, and EC2 instances that were created in the Multiple VPCs lab. You will set up VPC Flow logs for VPC A, generate some traffic, and then view the logs in CloudWatch.

❗

If you are running this lab at an **AWS-led event**, your facilitator has already set the region for you. If you are running it **self-service**, simply use the region you are working in. In either case, the region shown in the screenshots may not match your environment — this will not cause any problems.

If you are running this lab in **your own AWS Account**, it is recommended for all lab resources to be created in us-east-1 region so that the screenshots match your environment. This is not mandatory.

Prerequisites

- ▶ If you have not completed the Multiple VPCs section...

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CloudWatch

Amazon CloudWatch is a metrics repository. AWS services send metrics into the repository, and you retrieve statistics based on those metrics.

In this section, you will review some of the Networking metrics available in CloudWatch, create a dashboard to view a set of metrics on one page and set up an alarm on a threshold breach.

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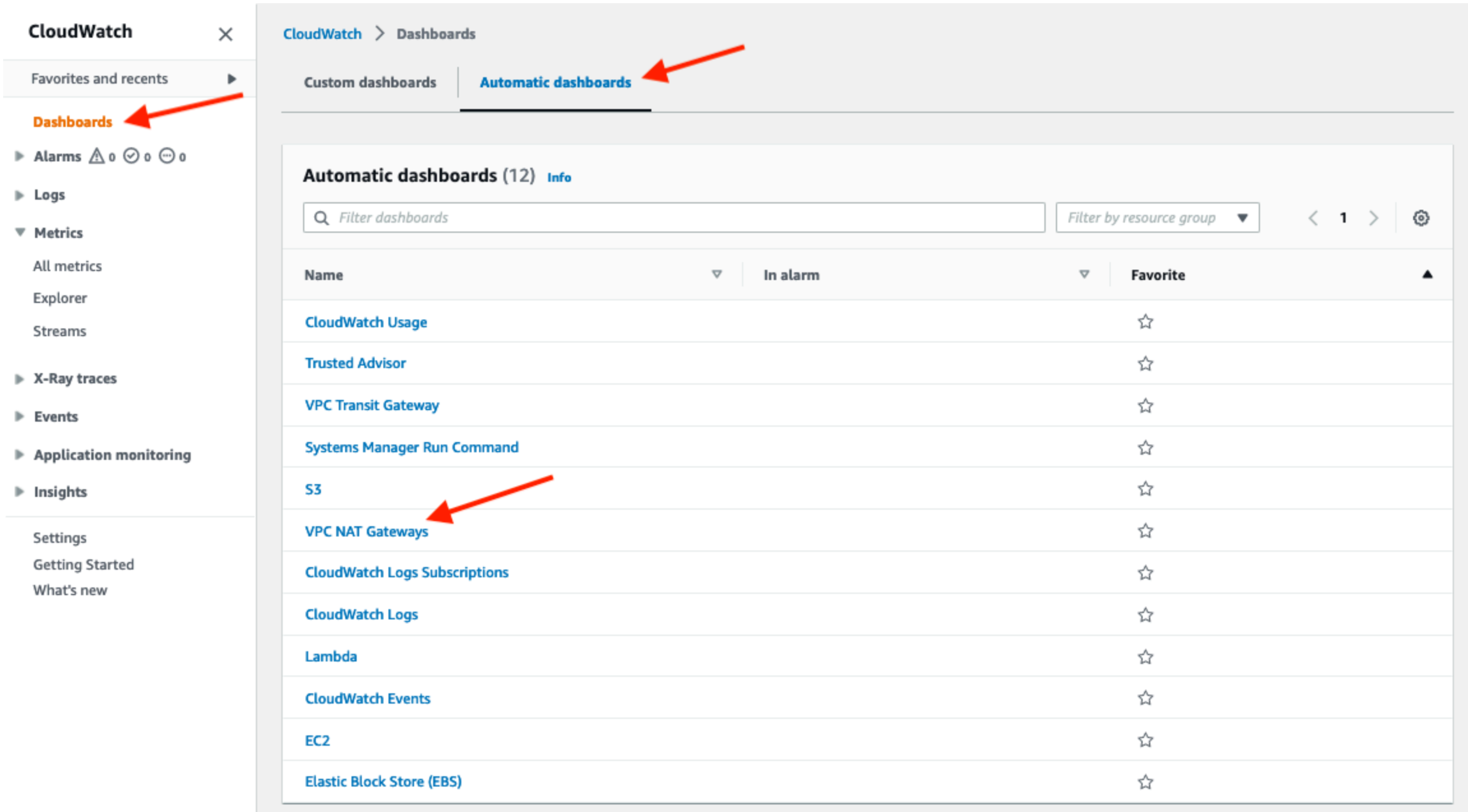
Language

English ▼

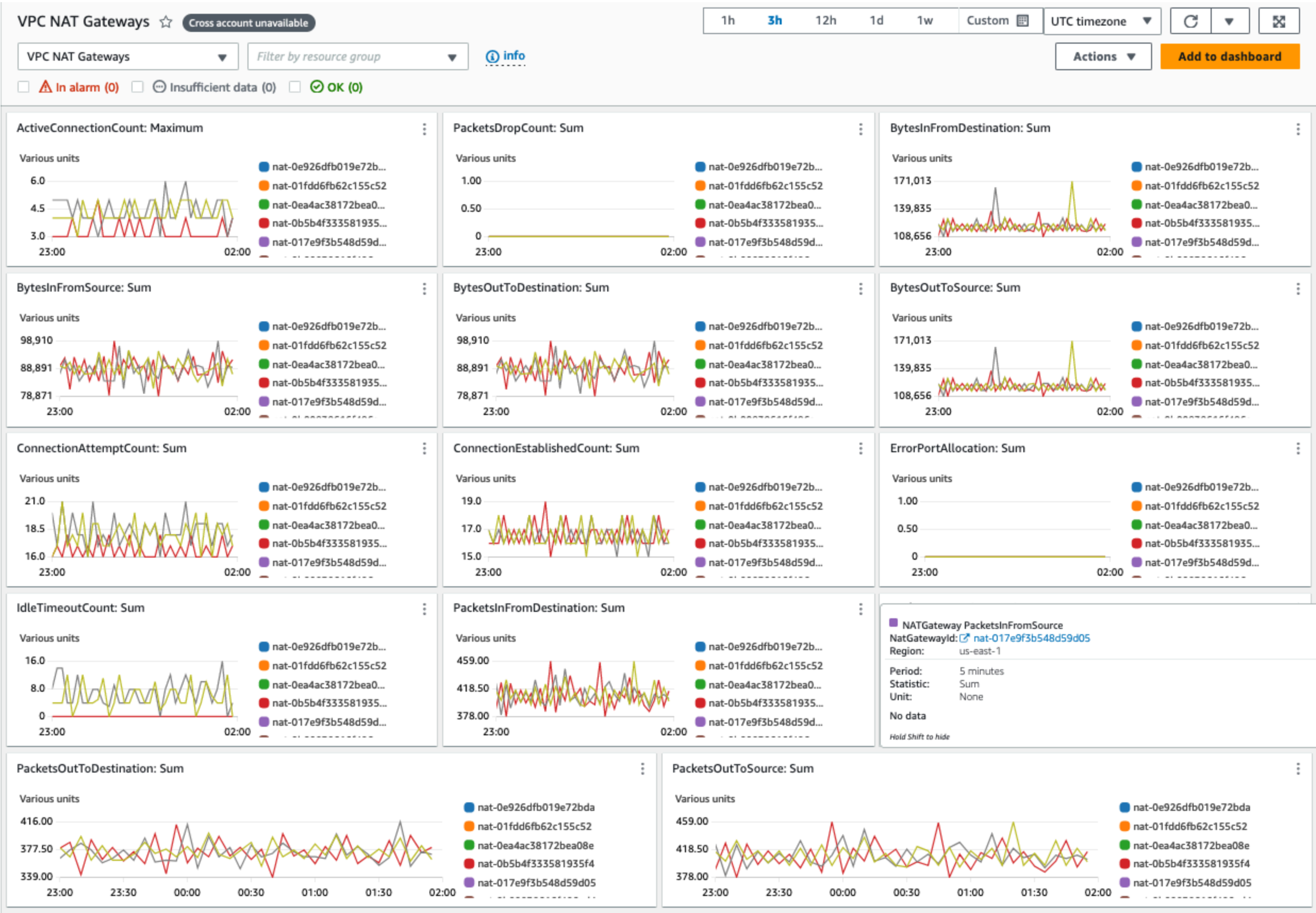
Review Automatic Dashboards

Amazon CloudWatch dashboards are customizable home pages in the CloudWatch console that you can use to monitor your resources in a single view, even those resources that are spread across different Regions. You can use CloudWatch dashboards to create customized views of the metrics and alarms for your AWS resources.

1. Navigate to the [CloudWatch console](#) 
2. Click **Dashboards**, then click **Automatic Dashboards** and **VPC NAT Gateways**



3. A dashboard will be shown that contains the key network metrics for NAT Gateways



4. Review the metrics

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Decline

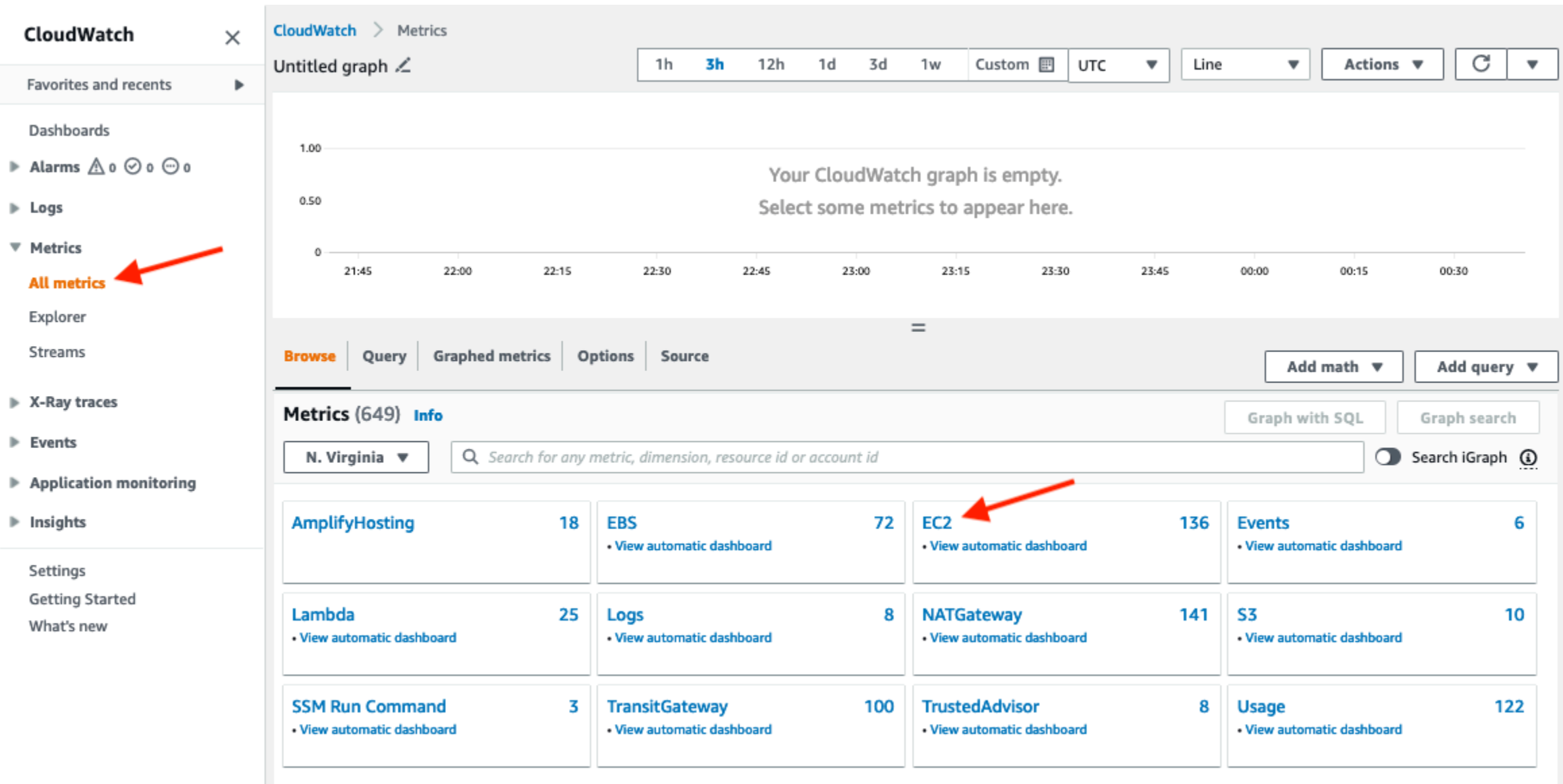
Customize

Create Custom Dashboard

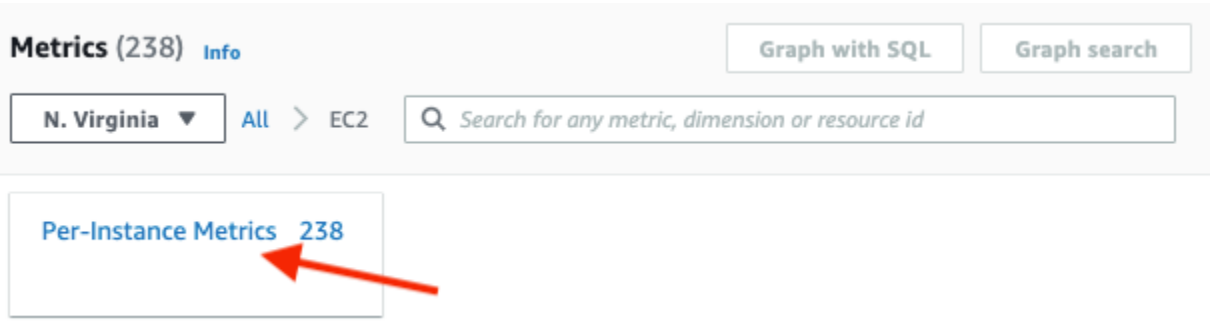
Amazon EC2 provides instance-level metrics that measure CPU, disk, and network performance. These metrics include bytes and packets in/out and collected by default and can be viewed in Amazon CloudWatch. Amazon EC2 has recently announced additional high-resolution instance level network performance metrics for Elastic Network Adapter (ENA). With these new metrics you can gain insights into traffic drops when network allowances are exceeded. For more information, please see [Amazon EC2 instance-level network performance metrics uncover new insights](#) 

In this section we will review the Network metrics in CloudWatch for the **VPC A Private AZ1 Server** instance in VPC A and create a custom dashboard with a widget showing the **NetworkIn** and **NetworkOut** metrics and another widget for the **NetworkPacketsIn** and **NetworkPacketsOut** metrics.

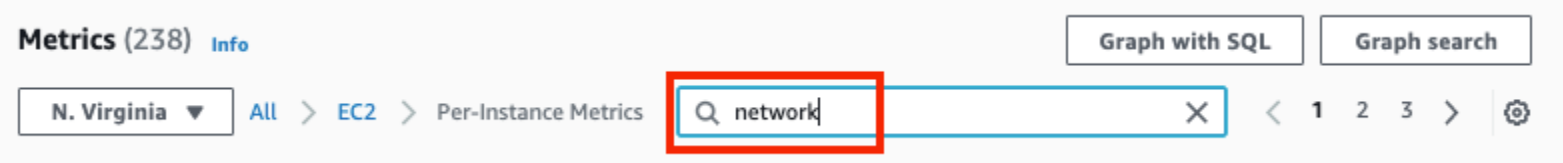
1. Navigate to the [CloudWatch console](#)
2. Under the **All Metrics** tab in the main screen click on **EC2**



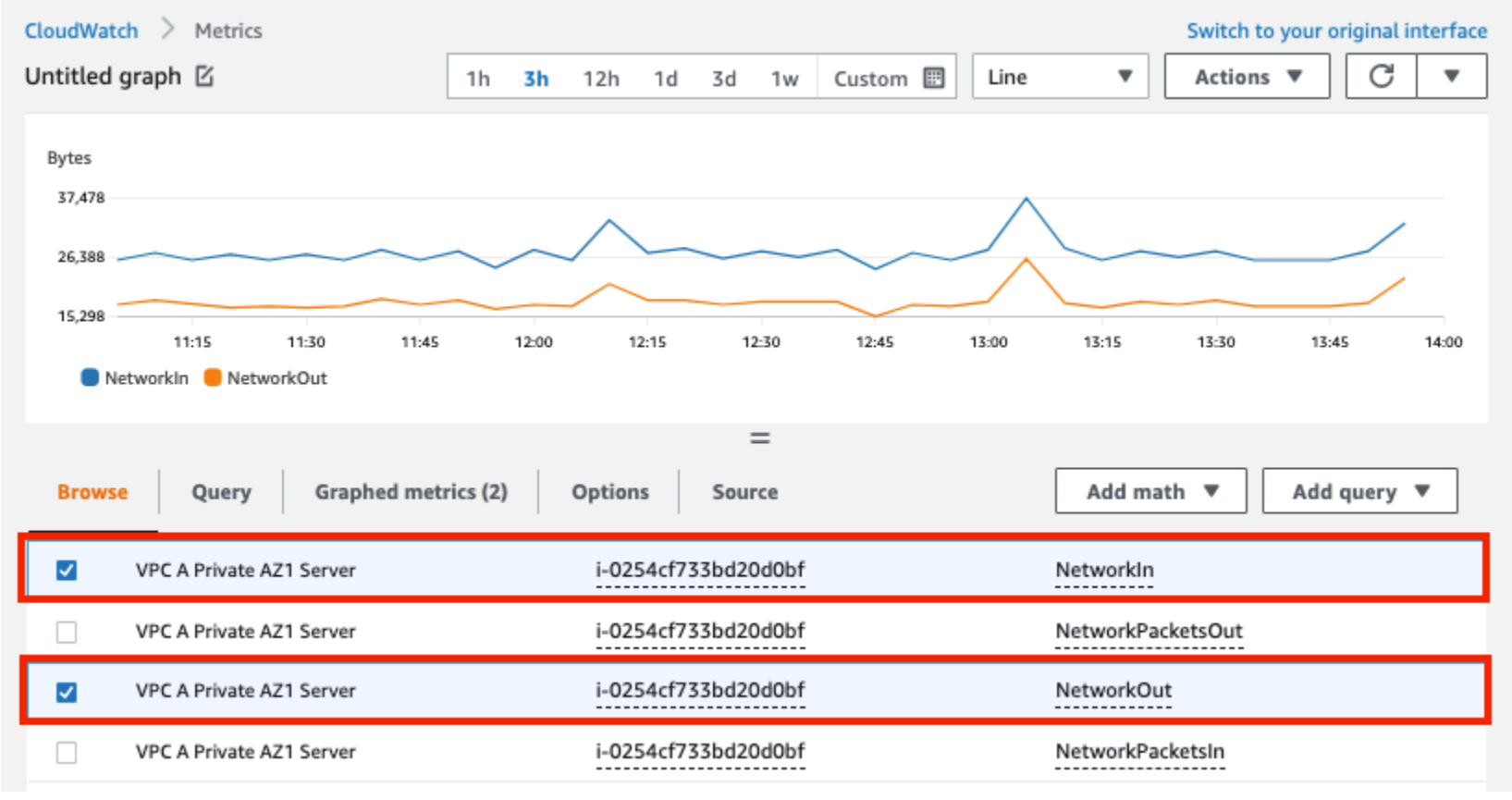
3. Click on **Per-Instance Metrics**.



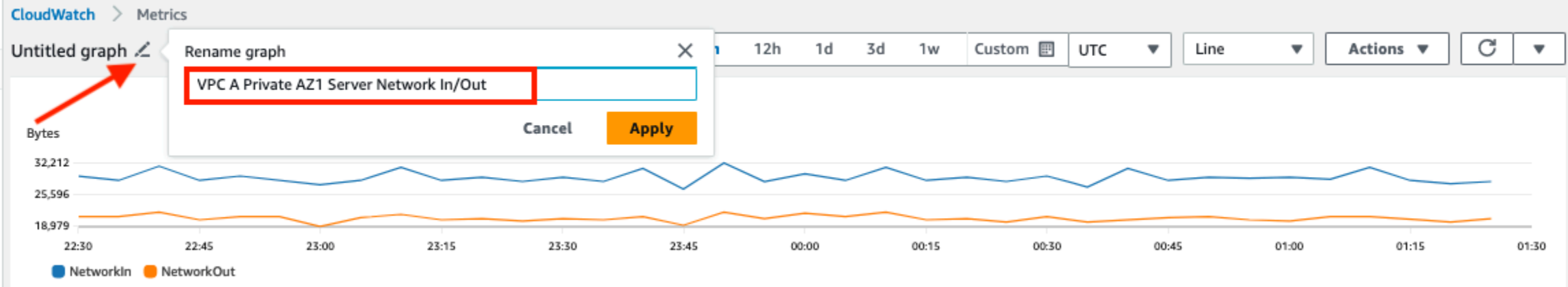
4. Filter the list for metrics with network in the name by entering **network** in the search



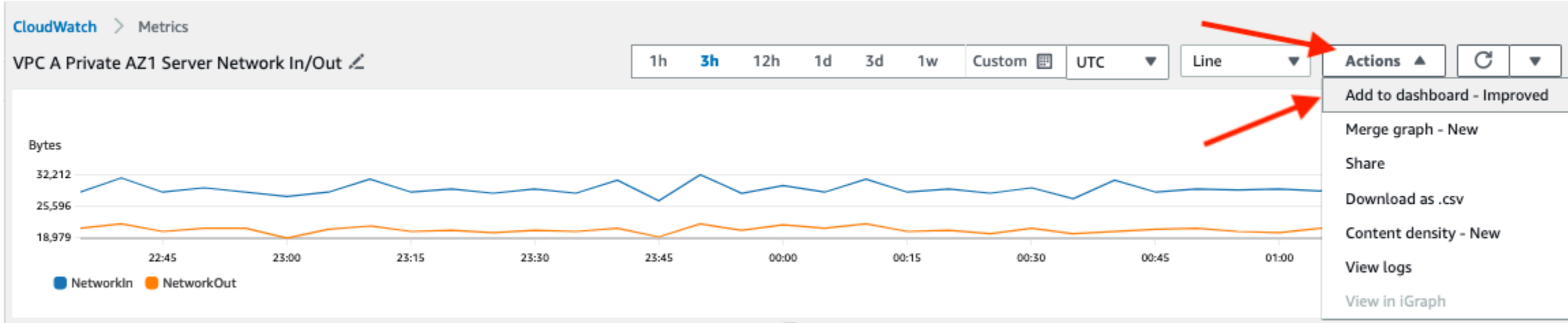
5. Select the metric **NetworkIn** and **NetworkOut** for the VPC A Private AZ1 Server



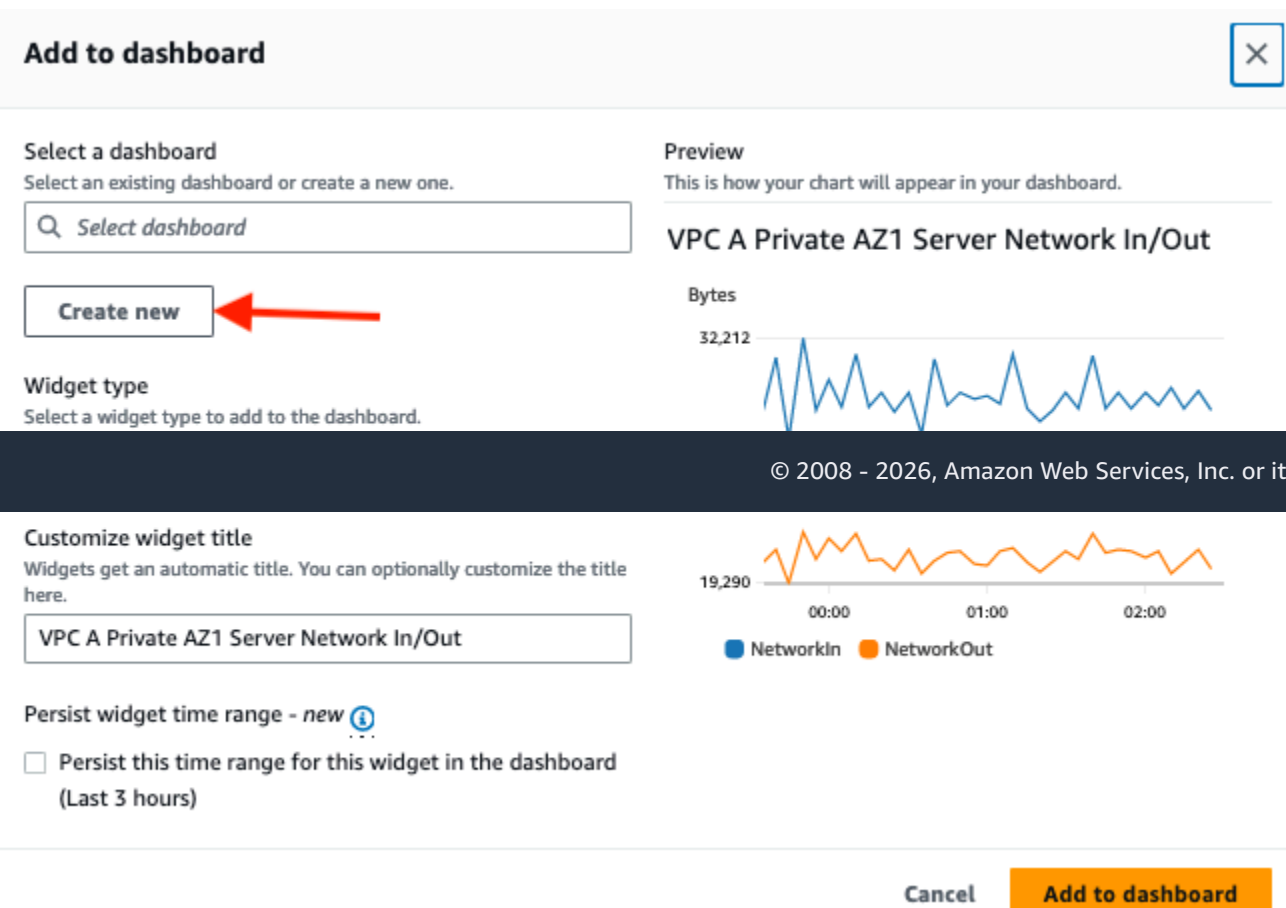
6. Click on the pencil icon at the top left corner of the graph and enter a name of VPC A Private AZ1 Server Network In/Out



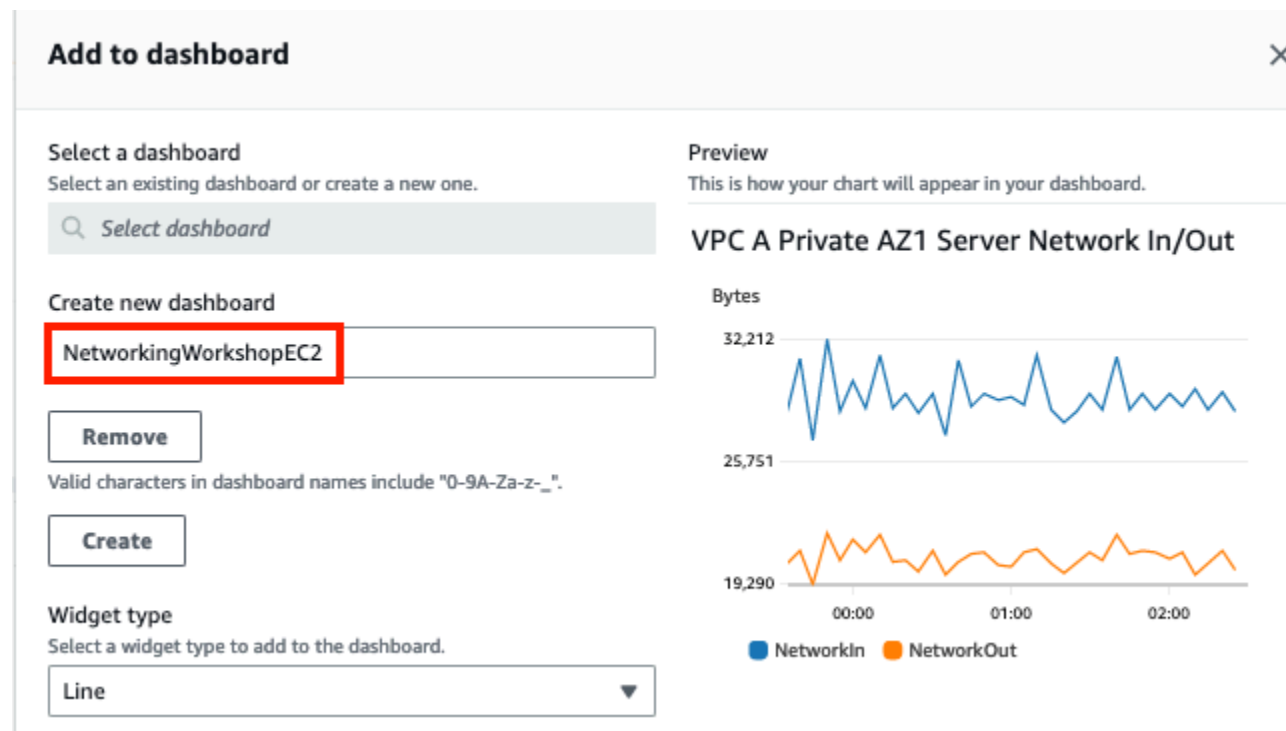
7. Click on **Actions** and select **Add to dashboard** from the dropdown



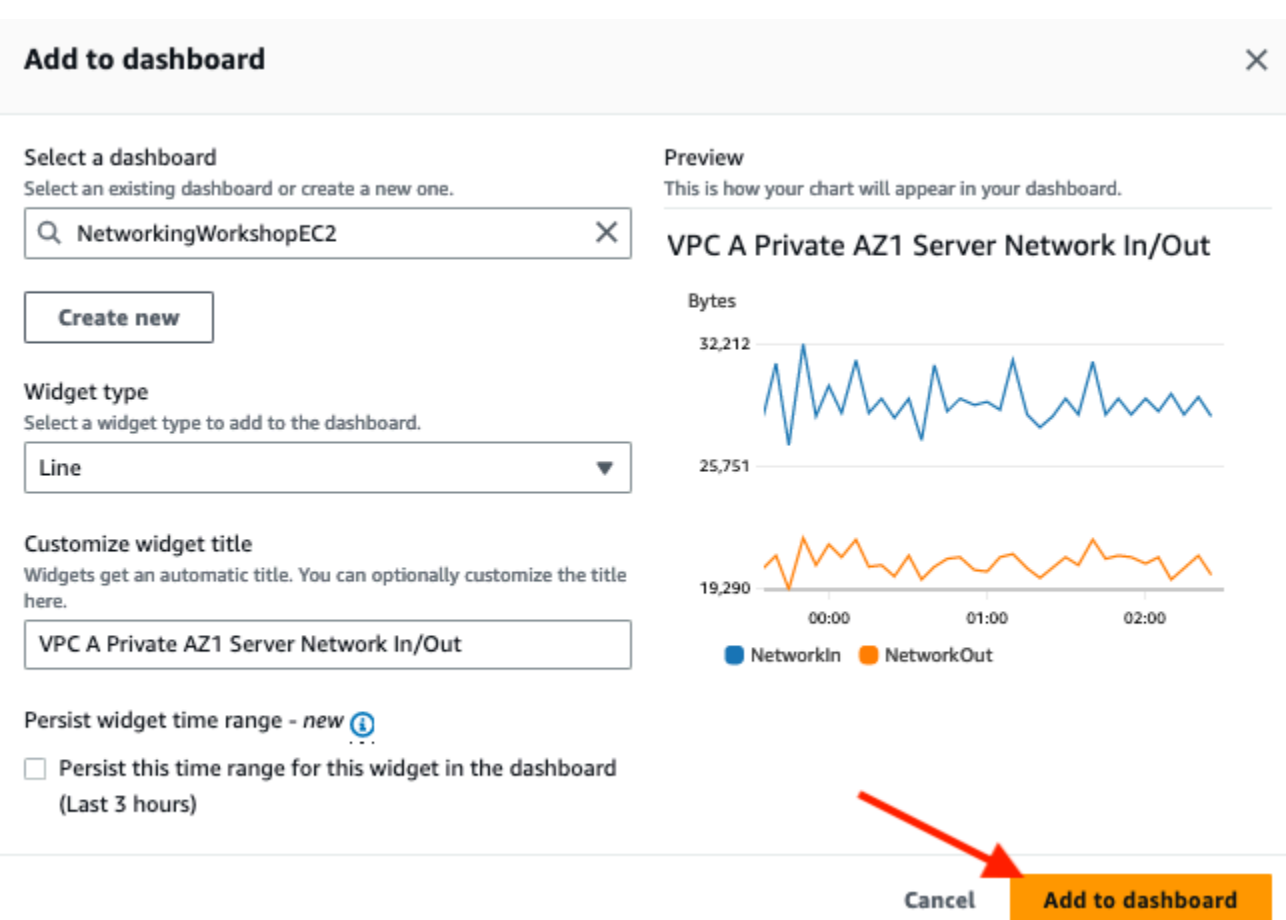
8. Click on **Create new**



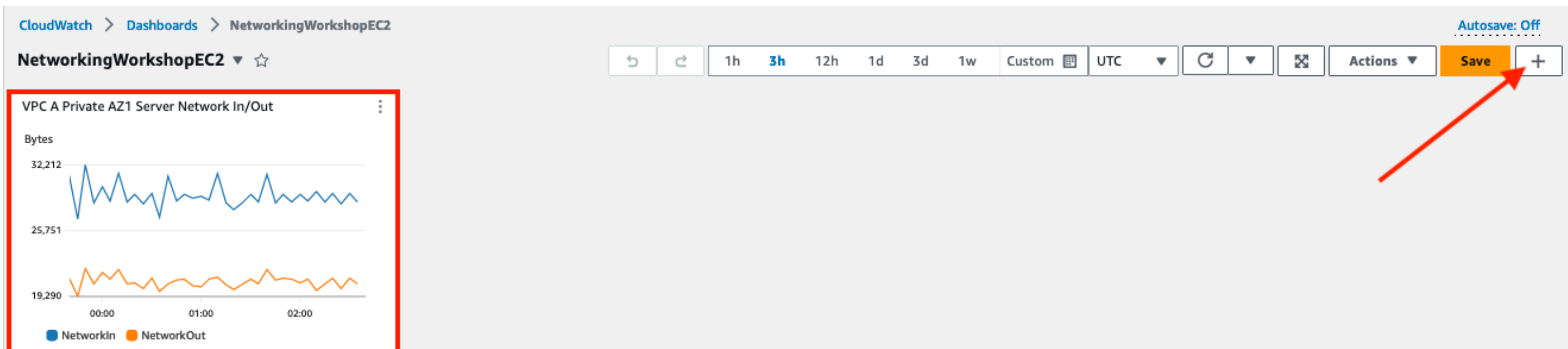
9. Enter `NetworkingWorkshopEC2` and click **Create**



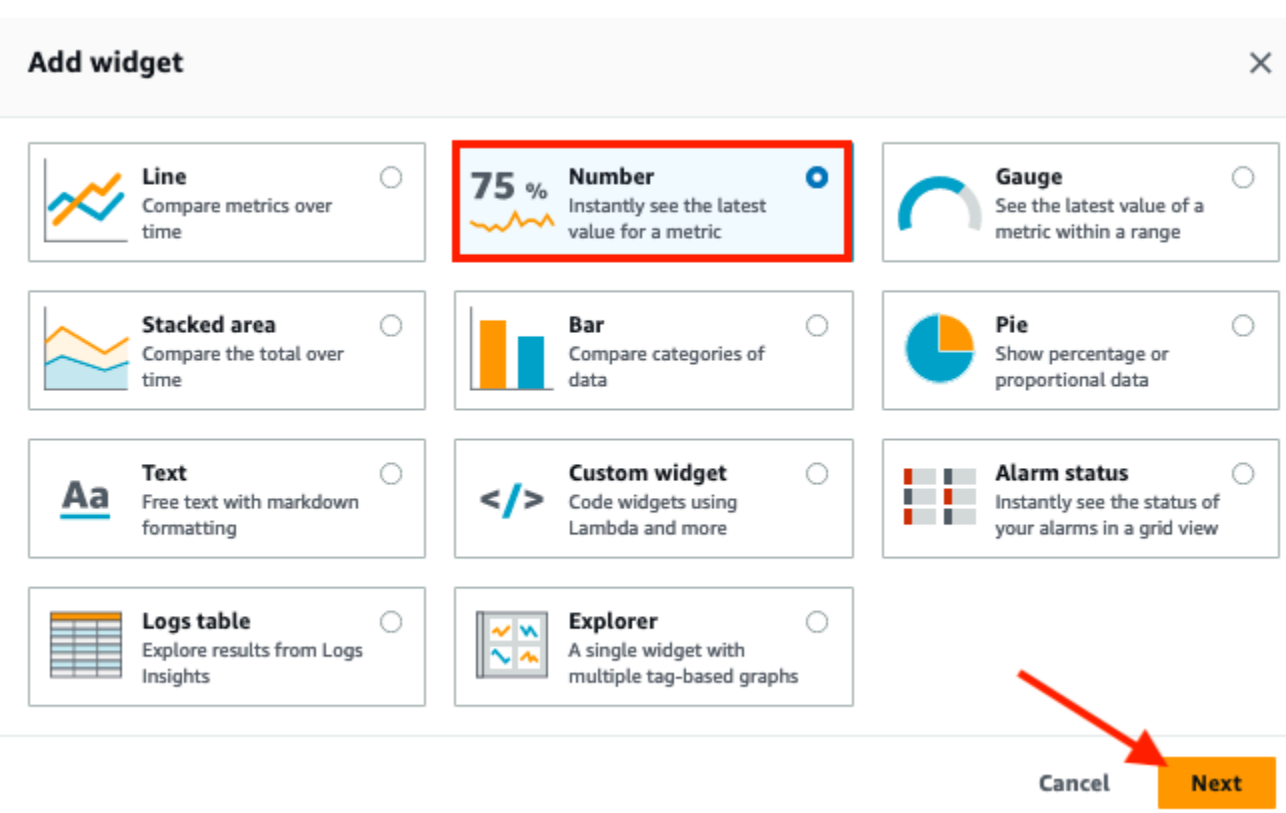
10. Click **Add to dashboard**



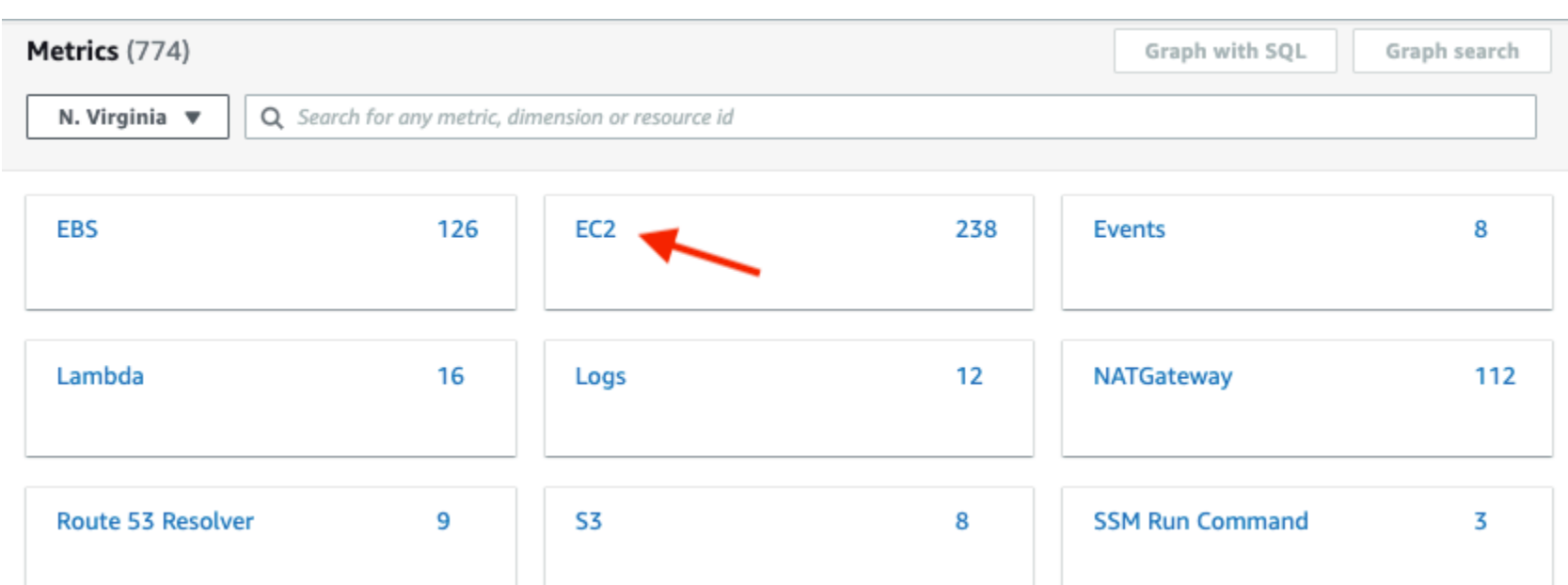
11. The graph of Network In/Out statistics in bytes for the EC2 instance will be added to a new dashboard. To add Packets In/Out statistics click on the + in the top right corner to add another widget



12. In the **Add Widget** screen select **Number**



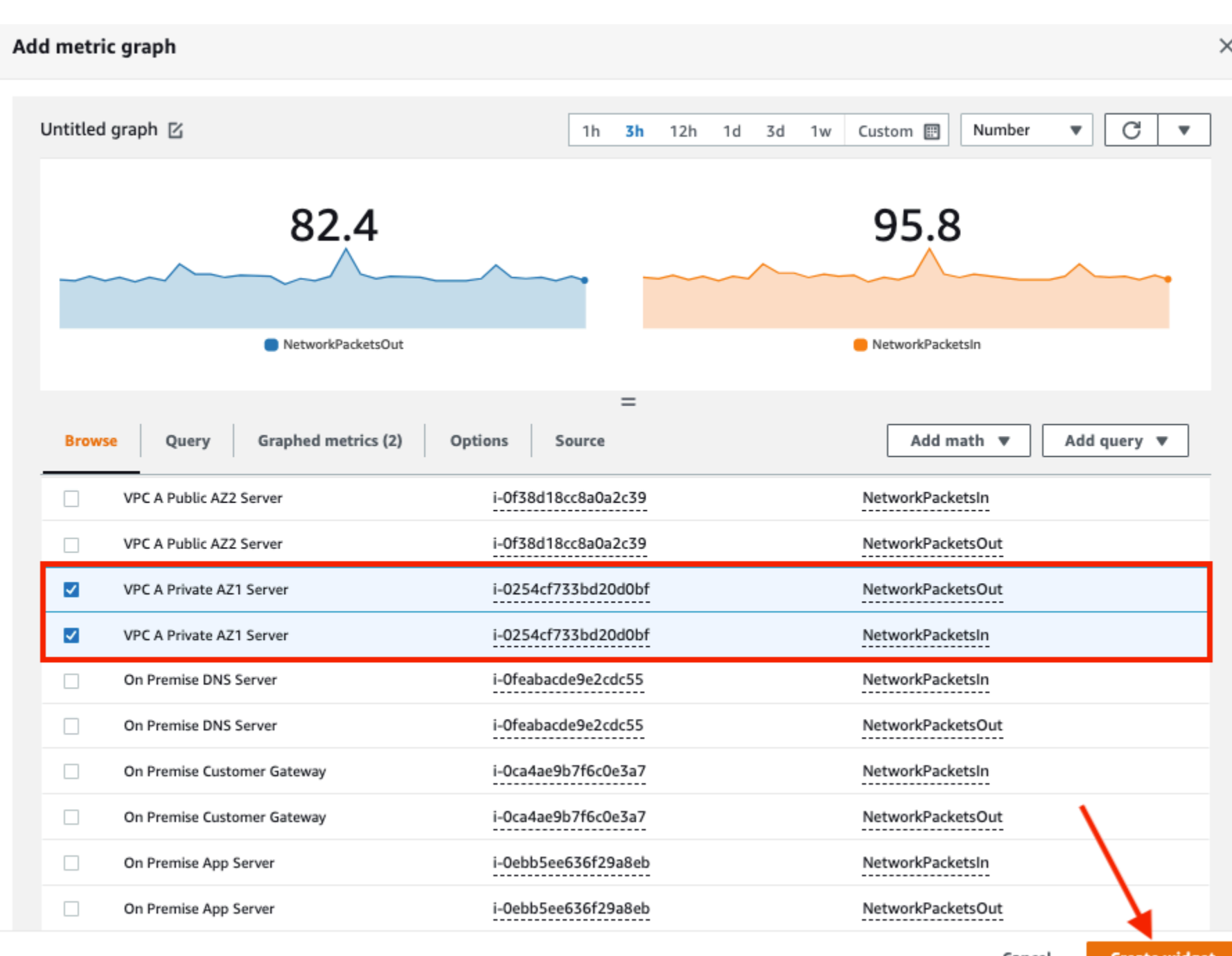
13. Click on **EC2**



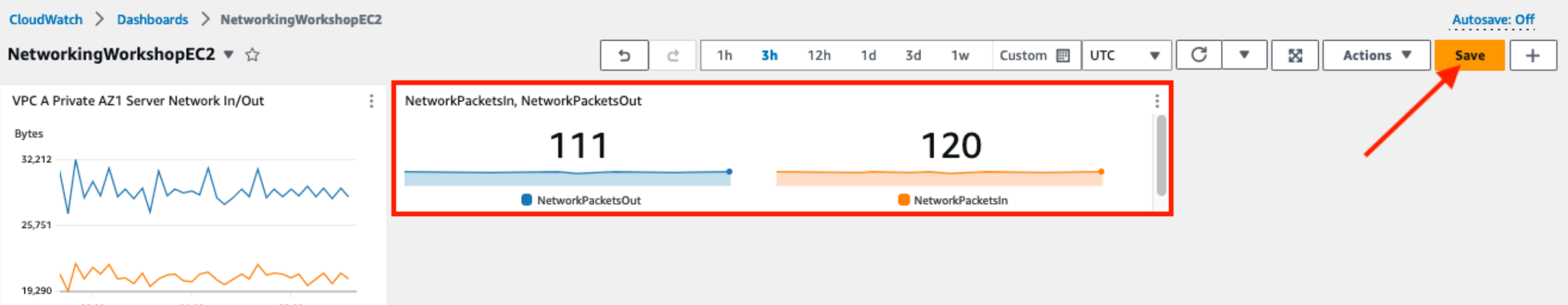
14. Click on **Per-Instance Metrics**



15. Enter `NetworkPackets` in the Search box and select the two lines for `vpc-4-private-az1-Subnet` and click **Create Widget**.



- 16 The new widget will then be placed on your dashboard. click **Save** to persist the dashboard



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Accepted

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VPC Flow Logs

VPC Flow Logs is a feature that enables you to capture information (metadata) about the IP traffic going to and from network interfaces in your VPC. For example, if you have a content delivery platform, flow logs can profile, analyze, and predict customer patterns of the content access, and track down top talkers and malicious calls.

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AWS Networking Workshop

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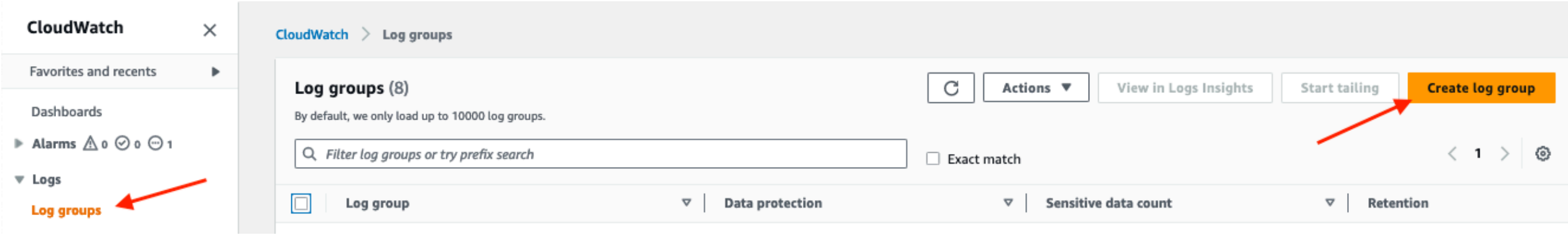
Create Flow Log

Create a flow log for all traffic in VPC A and save it to the destination log group **NetworkingWorkshopFlowLog**.

Create CloudWatch Log group

First, let's create a CloudWatch Log group to store flow logs in:

1. Navigate to the [CloudWatch Console](#)
2. In the menu on the left, click **Log groups** under **Logs** and click **Create log group** in the top right



3. Enter NetworkingWorkshopFlowLogsGroup as Log group name and click **Create**

Log group details

Log group name

NetworkingWorkshopFlowLogsGroup

Retention setting

Never expire

KMS key ARN - optional

Tags

A tag is a label that you assign to an Amazon Web Services resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your Amazon Web Services costs.

No tags are associated with this log group.

Add new tag

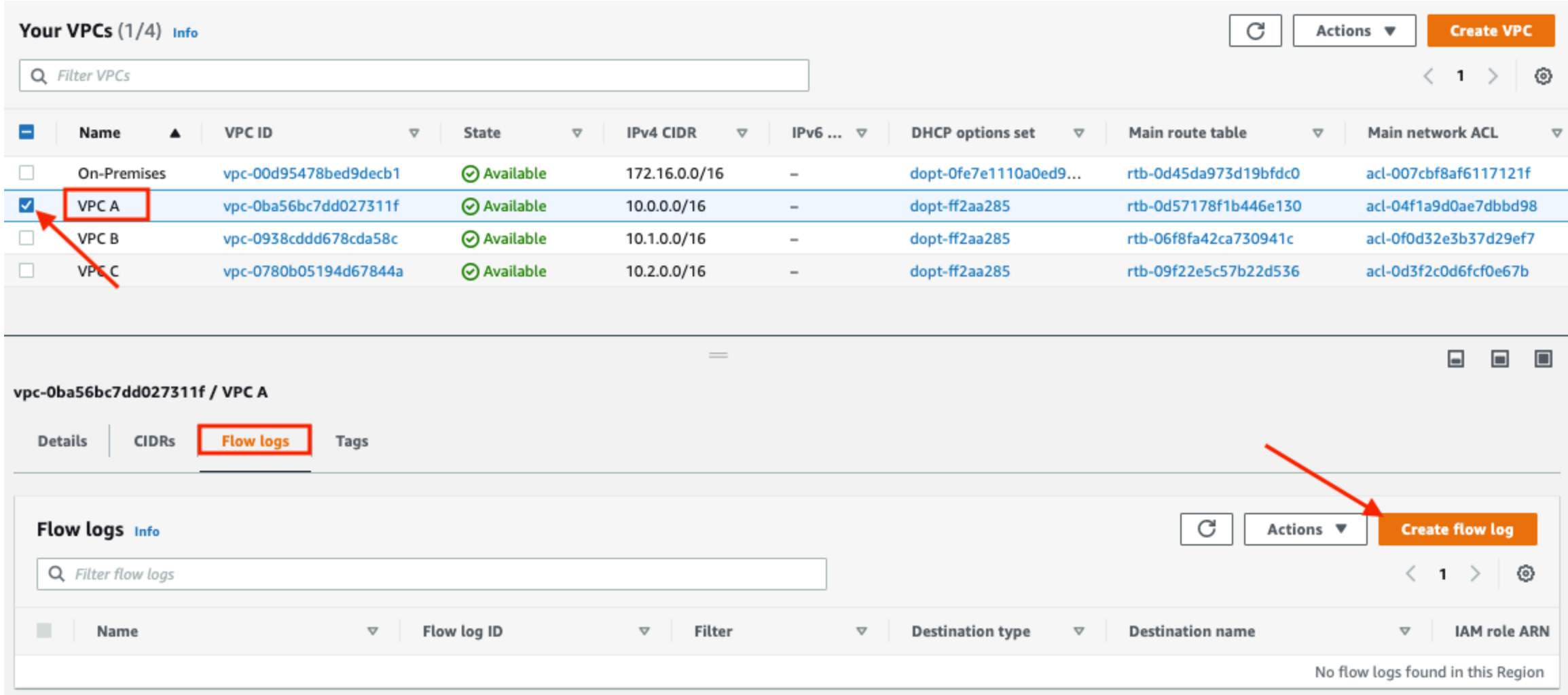
You can add up to 50 more tag(s).

Cancel

Create

Create VPC Flow log

4. In the VPC Dashboard navigate to [Your VPCs](#)
5. Select vpc_A, scroll down to the **Flow Logs** tab and click on **Create flow log**



6. In the resulting **Flow log settings** section

- Enter NetworkingWorkshopFlowLog in the **Name** field
- Set **Filter** to All
- Set **Maximum aggregation interval** to 1 minute
- Leave **Destination** as Send to CloudWatch Logs
- Set **Destination log group** as NetworkingWorkshopFlowLogsGroup
- Select NetworkingWorkshopFlowLogsRole from the **IAM role** dropdown (this IAM role was created by the workshop's base CloudFormation template)

Flow log settings

Name - optional

NetworkingWorkshopFlowLog

Filter

The type of traffic to capture (accepted traffic only, rejected traffic only, or all traffic).

Accept

Reject

All

Maximum aggregation interval

The maximum interval of time during which a flow of packets is captured and aggregated into a flow log record.

10 minutes

1 minute

Destination

The destination to which to publish the flow log data.

Send to CloudWatch Logs

Send to an Amazon S3 bucket

Destination log group

The name of the Amazon CloudWatch log group to which the flow log is published. A new log stream is created for each monitored network interface.

NetworkingWorkshopFlowLogsGroup

IAM role

The IAM role that has permission to publish to the Amazon CloudWatch log group. Set up permissions

NetworkingWorkshopFlowLogsRole

Log record format

Specify the fields to include in the flow log record.

AWS default format

Custom format

Format preview

Format preview

Copy

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7. Click on **Create flow log**

The completed setup of the VPC Flow logs.

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Accept

Decline

Customize

Select your cookie preferences

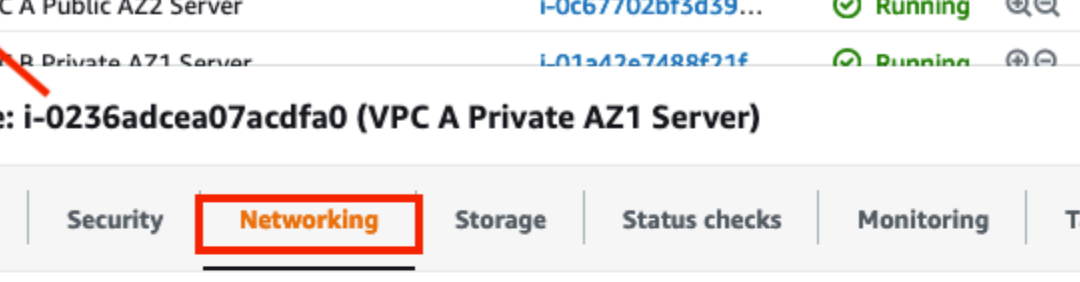
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View Flow Logs in CloudWatch

When publishing to CloudWatch, flow log data is published to a log group, and each network interface has a unique log stream in the log group. Log streams contain flow log records. You can create multiple flow logs that publish data to the same log group.

1. In the EC2 Dashboard, navigate to [Instances](#)
2. Select the *checkbox* next to VPC A Private AZ1 Server, scroll down to the **Networking** tab and make a note of the **Interface ID** under **Network Interfaces**



Instance: i-0236adcea07acdfa0 (VPC A Private AZ1 Server)

Details | **Security** | **Networking** | Storage | Status checks | Monitoring | Tags

► Networking details [Info](#)

▼ Network Interfaces [Info](#)

Network interfaces (1)

Filter network interfaces

Interface ID	Description	IPv4 Prefixes	IPv6 Prefixes
 eni-074cd8259ed...	-	-	-

VPC Flow logs can be sent to either an Amazon S3 bucket or CloudWatch. In this lab, you configured the flow logs from VPC A to be sent to CloudWatch.

3. Navigate to Log Groups in the [CloudWatch console](#) and click on the `NetworkingWorkshopFlowLogsGroup` log group

The screenshot shows the AWS CloudWatch 'Log groups' page. In the left-hand navigation pane, the 'Logs' section is expanded, and 'Log groups' is highlighted with a red arrow. The main content area shows a list of log groups. The first log group, 'NetworkingWorkshopFlow.logsGroup', is highlighted with a red arrow. The table columns include 'Log group', 'Data protection', 'Sensitive data co...', 'Retention', 'Metric filters', and 'Contributor Insig...'. The retention policy for the selected log group is 'Never expire'.

4. Click on the log stream matching the interface ID noted in step (1) to see the flow records for that interface (make sure to select the ENI from VPC A EC2)

NetworkingWorkshopFlowLogsGroup

Actions

View in Logs Insights

Search log group

▼ Log group details

Retention

Never expire

Creation time

39 minutes ago

Stored bytes

-

ARN

arn:aws:logs:us-east-1:5:.....log-group:NetworkingWorkshopFlowLogsGroup:*

KMS key ID

-

Metric filters

0

Subscription filters

0

Contributor Insights rules

-

Log streams

Metric filters

Subscription filters

Contributor Insights

Tags

Log streams (1/7)

↺

Delete

Create log stream

Search all

🔍

Filter log streams or try prefix search

<

1

>

⚙️

☐	Log stream	Last event time
☐	eni-0c44edfeaf2ecb3d-all	2022-06-30 13:08:23 (UTC+10:00)
☒	eni-053a823a73fde4246-all	2022-06-30 13:08:03 (UTC+10:00)
☐	eni-060bbec25e316395c-all	2022-06-30 13:07:59 (UTC+10:00)
☐	eni-06f5fd8d790f33964-all	2022-06-30 13:07:44 (UTC+10:00)
☐	eni-00410c37a9ee490a7-all	2022-06-30 13:07:44 (UTC+10:00)
☐	eni-054ef7a440a6b9331-all	2022-06-30 13:07:30 (UTC+10:00)
☐	eni-0C211b279f906e935-all	2022-06-30 13:07:00 (UTC+10:00)

5. Click on any entry to expand the log line

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events.
[Learn more about filter patterns](#)

☐ View as text

Actions

Create Metric Filter

Clear

1m

30m

1h

12h

Custom

Timestamp	Message	
No older events at this moment. Retry		
2022-06-30T13:08:03.000+10:00	2 506	eni-053a823a73fde4246 38.229.53.9 10.0.1.100 123 49324 17 1 76 1656558483 1656558527 ACCEPT OK
2 50	eni-053a823a73fde4246	38.229.53.9 10.0.1.100 123 49324 17 1 76 1656558483 1656558527 ACCEPT OK
2022-06-30T13:08:03.000+10:00	2 5	eni-053a823a73fde4246 10.0.1.100 38.229.53.9 49324 123 17 1 76 1656558483 1656558527 ACCEPT OK
2022-06-30T13:08:03.000+10:00	2 5	eni-053a823a73fde4246 52.46.156.29 10.0.1.100 443 44930 6 4 160 1656558483 1656558527 ACCEPT OK
2022-06-30T13:08:03.000+10:00	2 5	eni-053a823a73fde4246 10.0.1.100 52.46.156.29 44930 443 6 4 160 1656558483 1656558527 ACCEPT OK
2022-06-30T13:08:03.000+10:00	2 5	eni-053a823a73fde4246 168.235.86.33 10.0.1.100 123 51535 17 1 76 1656558483 1656558527 ACCEPT OK
2022-06-30T13:08:03.000+10:00	2 5	eni-053a823a73fde4246 10.0.1.100 168.235.86.33 51535 123 17 1 76 1656558483 1656558527 ACCEPT OK
2022-06-30T13:08:03.000+10:00	2 5	eni-053a823a73fde4246 52.46.143.243 10.0.1.100 443 52078 6 25 7251 1656558483 1656558527 ACCEPT OK
2022-06-30T13:08:03.000+10:00	2 5	eni-053a823a73fde4246 10.0.1.100 52.46.143.243 52078 443 6 22 5327 1656558483 1656558527 ACCEPT OK
2022-06-30T13:08:03.000+10:00	2 5	eni-053a823a73fde4246 52.46.146.146 10.0.1.100 443 34868 6 26 7291 1656558483 1656558527 ACCEPT OK
2022-06-30T13:08:03.000+10:00	2 5	eni-053a823a73fde4246 10.0.1.100 52.46.146.146 34868 443 6 22 5327 1656558483 1656558527 ACCEPT OK
2022-06-30T13:08:03.000+10:00	2 5	eni-053a823a73fde4246 52.46.143.243 10.0.1.100 443 39966 6 26 7291 1656558483 1656558527 ACCEPT OK
2022-06-30T13:08:03.000+10:00	2 5	eni-053a823a73fde4246 10.0.1.100 52.46.143.243 39966 443 6 22 5327 1656558483 1656558527 ACCEPT OK
2022-06-30T13:08:03.000+10:00	2 5	eni-053a823a73fde4246 209.54.183.5 10.0.1.100 443 59234 6 2 160 1656558483 1656558527 ACCEPT OK

Field	Value
VPC Flow Logs version	Version: 2
AWS account ID for the flow log	Account ID: 379063898357
Source and destination IPv4/IPv6 address	Interface ID: eni-04b10a1942977452f Source Address: 172.16.254.34 Destination Address: 32.68.32.56
IANA protocol number of the traffic	Source Port: 36490 Destination Port: 443 Protocol: 6
ID of the elastic network interface for which traffic is recorded	Packets: 77
Source and destination port	
Number of packets/bytes transferred during the capture window	
the capture window	Start: 1560385064 End: 1560385070
Status of the log: OK, NODATA, or	Action: ACCEPT Log Status: OK
Action of the traffic: ACCEPT or REJECT based on the security group or networking	

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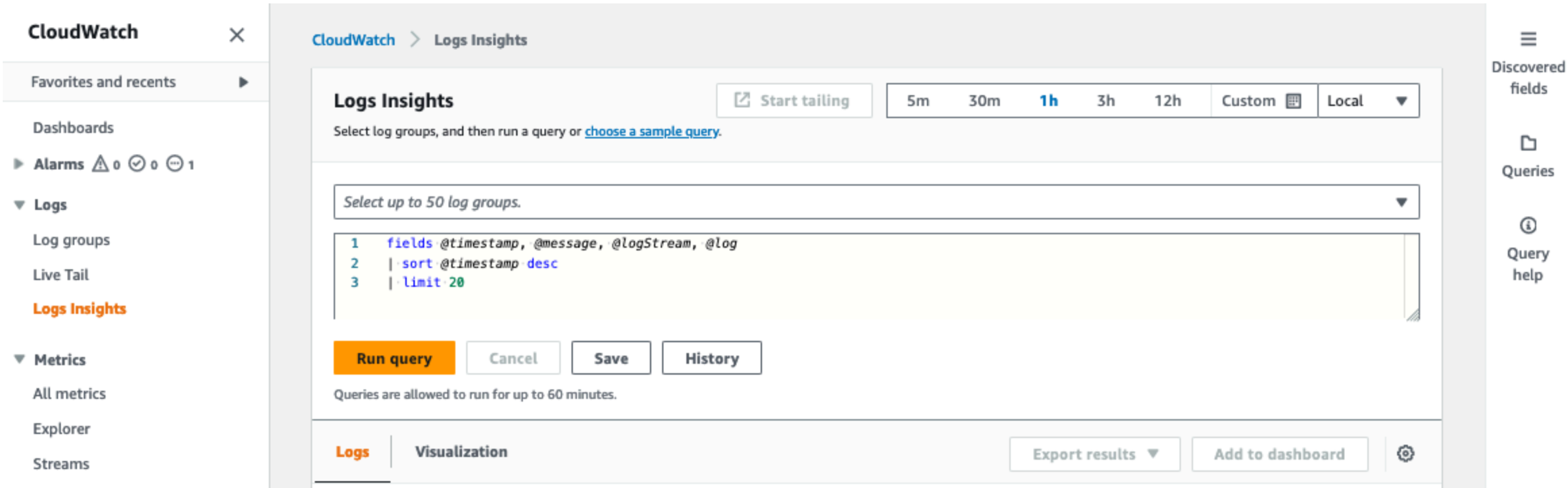
Language

English

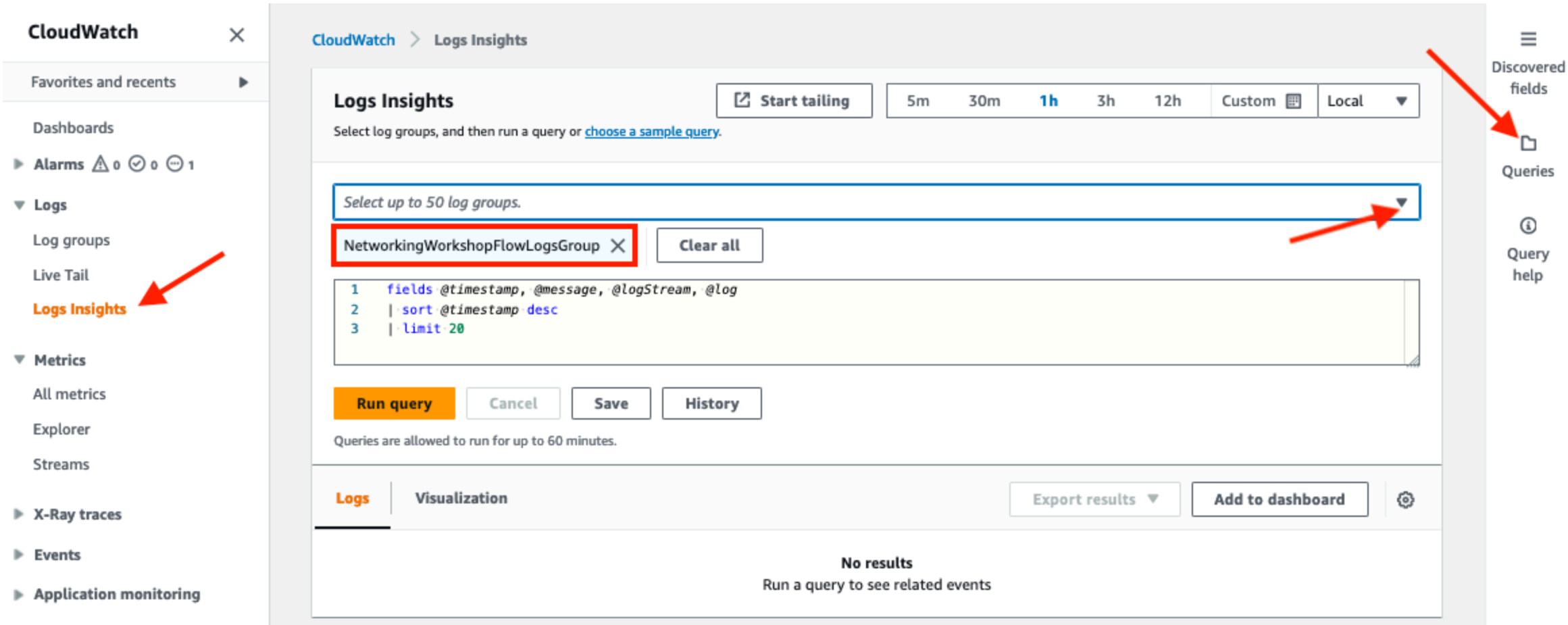
Query Flow Log for Insights

CloudWatch Logs Insights enables you to interactively search and analyze log data in CloudWatch Logs, including VPC flow logs. You can perform queries to help you more efficiently and effectively respond to operational issues. In the section you will run a query to show the top 10 talkers based on bytes transferred.

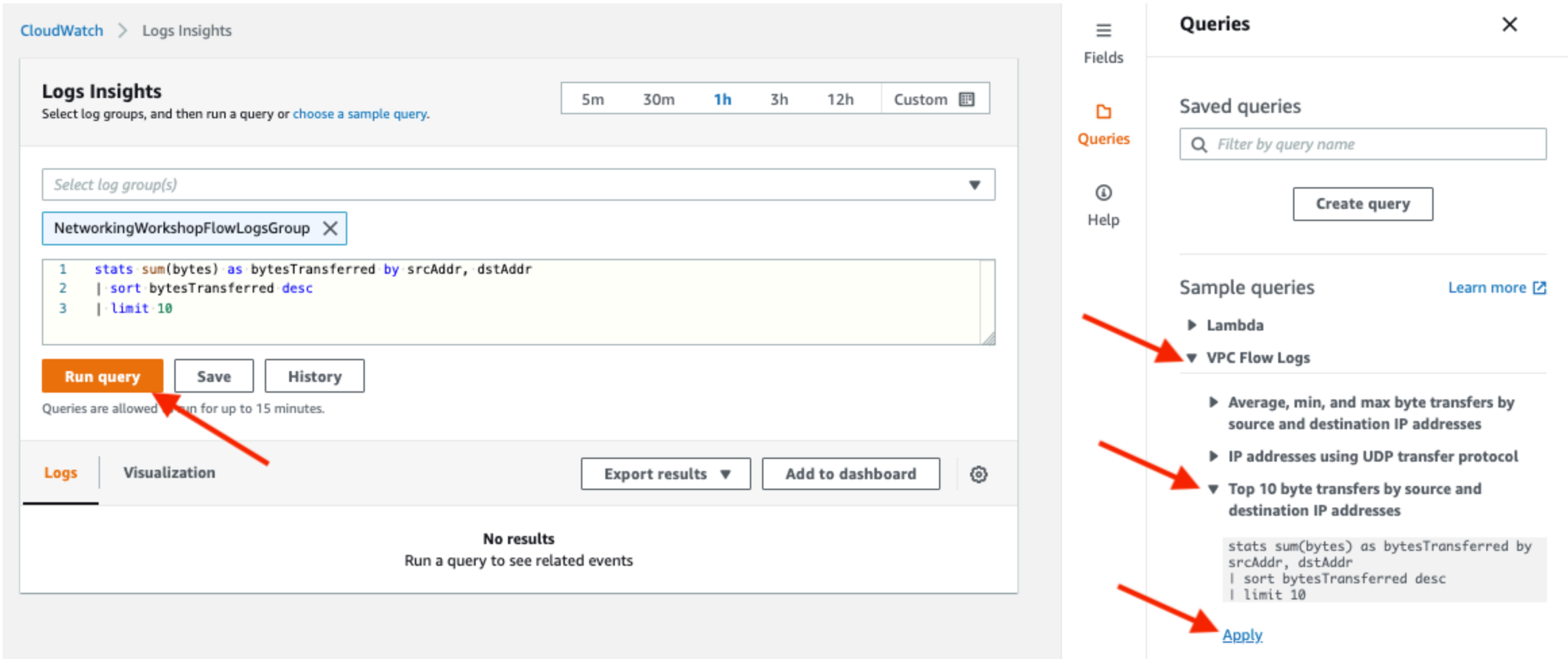
1. In the [CloudWatch console](#) click on **Logs Insights**



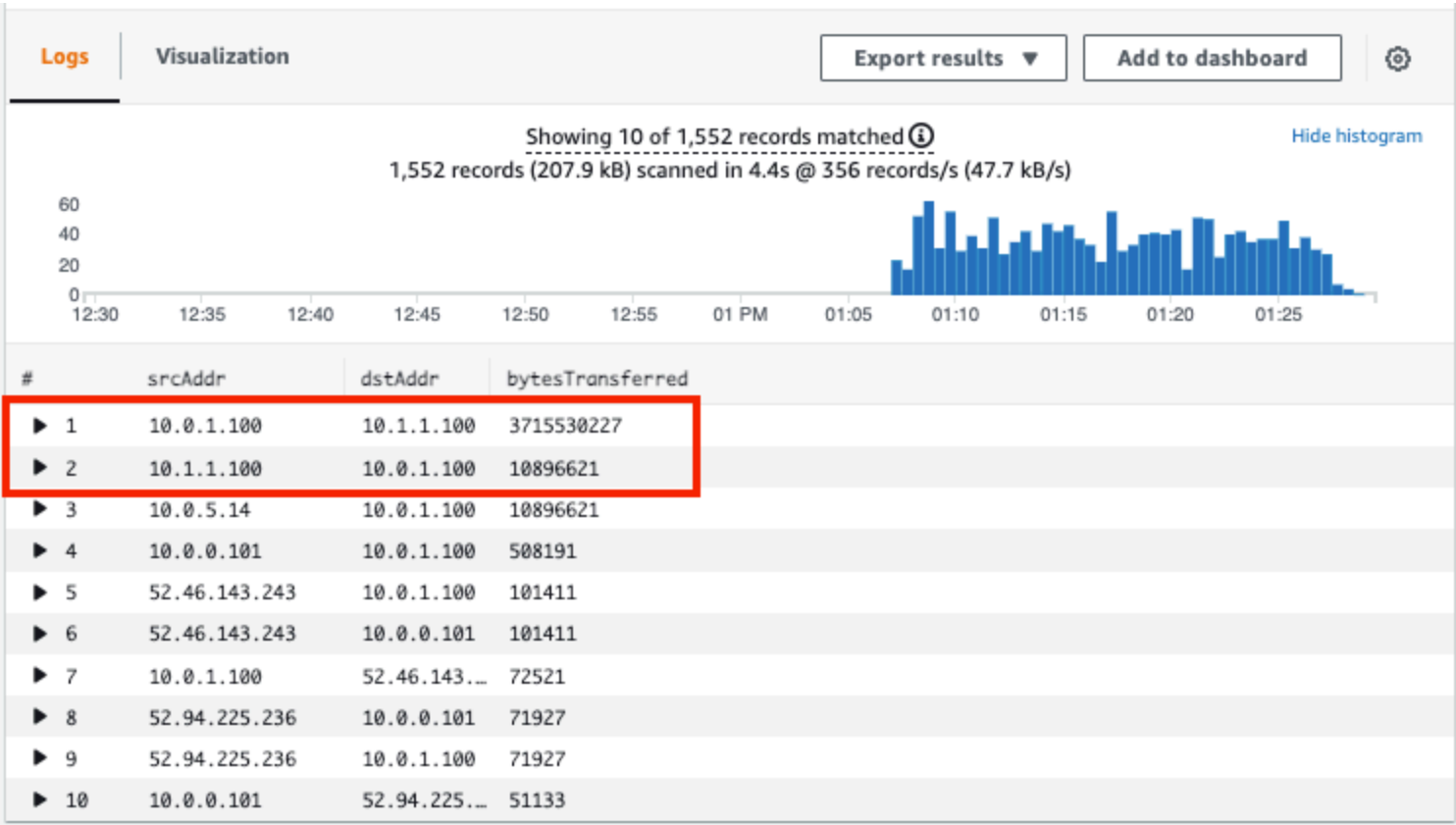
2. Select `NetworkingWorkshopFlowLogsGroup` from the **Select log group(s)** dropdown and click the **Queries** folder on the right hand side



3. Click **Top 10 byte transfers by source and destination IP addresses** under **VPC Flow Logs**, click **Apply** and then **Run query**



4. Review the query results. Do you recognize the top two IP addresses?



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Check Alarm Notification

Generating traffic to the instance in VPC B via iperf should have triggered the alarm we created earlier.

Open the inbox for the email destination you configured for the alarm and check that an email notification has arrived.

ALARM: "NetworkingWorkshopAlarm" in US East (N. Virginia)



AWS Notifications <no-reply@sns.amazonaws.com>
To:

You are receiving this email because your Amazon CloudWatch Alarm "NetworkingWorkshopAlarm" in the US East (N. Virginia) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [6.800438427E9 (18/09/23 05:20:00)] was greater than the threshold (1000000.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Monday 18 September, 2023 05:25:51 UTC".

View this alarm in the AWS Management Console:
<https://us-east-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-1#alarmsV2:alarm/NetworkingWorkshopAlarm>

Alarm Details:

- Name:

NetworkingWorkshopAlarm

- Description:

- State Change:

INSUFFICIENT_DATA -> ALARM

- Reason for State Change:

Threshold Crossed: 1 out of the last 1 datapoints [6.800438427E9 (18/09/23 05:20:00)] was greater than the threshold (1000000.0) (minimum 1 datapoint for OK -> ALARM transition).

- Timestamp:

Monday 18 September, 2023 05:25:51 UTC

- AWS Account:

- Alarm Arn:

arn:aws:cloudwatch:us-east-1:::alarm:NetworkingWorkshopAlarm

Threshold:
- The alarm is in the ALARM state when the metric is GreaterThanThreshold 1000000.0 for at least 1 of the last 1 period(s) of 60 seconds.

Monitored Metric:

- MetricNamespace:

AWS/EC2

- MetricName:

NetworkIn

- Dimensions:

[InstanceId = i-05b5c2bf301f7440e]

- Period:

60 seconds

- Statistic:

Maximum

- Unit:

not specified

- TreatMissingData:

missing

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